Supplementary information for: A gas-transport model predicts the *Arabidopsis* spaceflight transcriptome, and illumination separates six flight contrasts

Richard Barker[1*]

[1]Independent researcher, [TODO], [TODO].

Corresponding author(s). E-mail(s): dr.richard.barker@gmail.com;

Supplementary Tables

Supplementary Figures

Table S1 Zero-dimensional (0-D) enclosure model specifications across hardware classes. Parameters governing dynamic mass balance, boundary-layer drawdown, and 12 h carbon gain.

Parameter	**BRIC (Sealed)**	**CARA (Tape)**	**VEGGIE (Vented)**
Headspace volume V	30 mL (30×10^{-6} m^3)	30 mL (30×10^{-6} m^3)	∞ (open/circulated)
Leaf area A_{leaf}	3 cm^2 (3×10^{-4} m^2)	3 cm^2 (3×10^{-4} m^2)	3 cm^2
Enclosure surface A_{dish}	28 cm^2 (2.8×10^{-3} m^2)	28 cm^2 (venting lid/sides)	N/A
Tape / vent permeance	0.0 mol m^{-2} s^{-1} (sealed)	0.006 mol m^{-2} s^{-1}	Vented to cabin
Air density ρ_{air} (25 °C)	41.6 mol m^{-3}	41.6 mol m^{-3}	41.6 mol m^{-3}
Initial C_a (CO_2)	400 μmol mol^{-1} (ppm)	400 μmol mol^{-1}	400 μmol mol^{-1}
Initial O_2	210 mmol mol^{-1} (21 %)	210 mmol mol^{-1}	210 mmol mol^{-1}
Boundary air condition	Zero flux ($\partial C/\partial n = 0$)	Relaxed to cabin	Pinned at cabin air
Boundary-layer g_{bl} (μg)	0.272 mol m^{-2} s^{-1} (rosette)	0.272 mol m^{-2} s^{-1}	0.272 mol m^{-2} s^{-1}
Mixing assumption	Well-mixed headspace	Well-mixed headspace	Actively swept
12 h photoperiod gain	1 % of Earth	90 % of Earth	100 % of Earth

Table S2 Farquhar–von Caemmerer–Berry (FvCB) C_3 model parameterization at 25 °C. Kinetic constants and activation energies from in-vivo Bernacchi et al. (2001, 2003); photosynthetic capacities are Arabidopsis mid-range values.

Symbol	Quantity	Value (25 °C)	Units / Source
$V_{c\,\max,25}$	Maximum carboxylation capacity	90.0	μmol m^{-2} s^{-1}
$J_{\max,25}$	Maximum electron transport rate	153.0	μmol m^{-2} s^{-1} ($\sim 1.7 \times V_{c\,\max}$)
$R_{d,25}$	Day respiration	1.35	μmol m^{-2} s^{-1}
$K_{c,25}$	Michaelis constant for CO_2	404.9	μmol mol^{-1} (Bernacchi 2001)
$K_{o,25}$	Michaelis constant for O_2	278.4	mmol mol^{-1} (Bernacchi 2001)
Γ^*_{25}	CO_2 compensation point w/o R_d	42.75	μmol mol^{-1} at 210 mmol mol^{-1} O_2
$S_{c/o,25}$	Rubisco specificity	2456.1	mol mol^{-1} ($0.5 \times O_{\mathrm{ref}}/\Gamma^*_{25}$)
g_s	Stomatal conductance to CO_2	0.20	mol m^{-2} s^{-1}
g_m	Mesophyll conductance to CO_2	0.30	mol m^{-2} s^{-1}
θ	Curvature of light response	0.70	Dimensionless
α	Quantum yield of electron transport	0.30	mol e$^-$ / mol photon
$E_a(V_{c\,\max})$	Activation energy $V_{c\,\max}$	65.33	kJ mol^{-1}
$E_a(J_{\max})$	Activation energy $J_{\max}$	43.90	kJ mol^{-1}
$E_a(R_d)$	Activation energy R_d	46.39	kJ mol^{-1}
$E_a(K_c)$	Activation energy K_c	79.43	kJ mol^{-1}
$E_a(K_o)$	Activation energy K_o	36.38	kJ mol^{-1}
$E_a(\Gamma^*)$	Activation energy Γ^*	37.83	kJ mol^{-1}

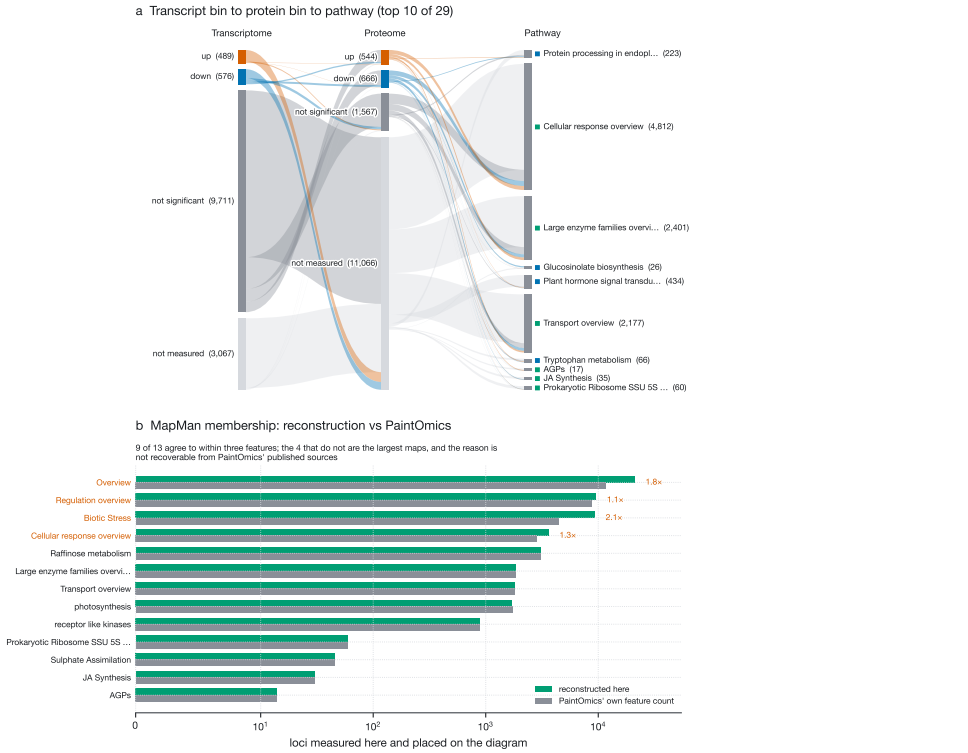
Table S3 NASA OSDR metabolomics census: all twelve metabolite-profiling studies. Queried 2026-09-08 across all 567 OSDR accessions. Demonstrates that 0 of 12 are plant studies.

Accession	Organism	Study Title / Focus
OSD-108	*Mus musculus*	Muscle atrophy and metabolic alteration in spaceflight
OSD-116	*Mus musculus*	Spaceflight-induced changes in liver metabolism
OSD-138	*Mus musculus*	Multi-omics of rodent eye and retina following flight
OSD-145	*Homo sapiens*	Human astronaut serum metabolomics (Twin Study)
OSD-343	*Mus musculus*	Cecal metabolome and microbiome in microgravity
OSD-360	*Rattus norvegicus*	Rat musculoskeletal metabolomics
OSD-417	*Mus musculus*	Rodent spleen and bone marrow metabolic profiles
OSD-571	*Microbial*	ISS environmental microbiome and metabolome
OSD-734	*Mus musculus*	Murine cardiovascular and plasma metabolomics
OSD-832	*Homo sapiens*	Commercial astronaut plasma metabolomic profiling
OSD-833	*Homo sapiens*	Spaceflight urine metabolomics time series
OSD-834	*Homo sapiens*	Astronaut peripheral blood mononuclear cell metabolome

Table S4 Specification and physiological rationale for the 21 predicted metabolite pools. $P_{\mathrm{flt}}/P_{\mathrm{gc}} = J_{\mathrm{flt}}/J_{\mathrm{gc}}$ under quasi-steady state. $\mathrm{Log_2FC}$ values are model predictions.

Compound	**KEGG**	**Driver**	**Tier**	**Physiological Basis / Rationale**
2-Phosphoglycolate	C00988	V_o	1	Primary product of Rubisco oxygenase activity
Glycolate	C00160	V_o	1	Dephosphorylated 2-PG exported to peroxisome
Glyoxylate	C00048	V_o	1	Peroxisomal oxidation of glycolate by GOX
Glycine	C00037	V_o	1	Transamination product; mitochondrial GDC substrate
L-Serine	C00065	V_o	1	Mitochondrial condensation product of 2 Gly by SHMT
Hydroxypyruvate	C00168	V_o	1	Peroxisomal transamination product of Serine
D-Glycerate	C00258	V_o	1	Reduced hydroxypyruvate; chloroplast kinase substrate
Ammonia	C00014	V_o	1	Stoichiometric release from GDC reaction in C_2 cycle
3-Phosphoglycerate	C00197	A	1	Stable product of Rubisco carboxylation
Sucrose	C00089	A	1	Primary transport carbohydrate synthesized from triose-P
D-Glucose	C00031	A	1	Free hexose pool downstream of sucrose cleavage
D-Fructose	C00095	A	1	Free hexose pool downstream of invertase activity
Maltose	C00208	A	1	Starch breakdown and export intermediate
Starch	C00369	A	1	Chloroplastic insoluble reserve from surplus assimilation
RuBP	C01182	$-C_c$	2	Accumulates upstream of CO_2-limited Rubisco
L-Asparagine	C00152	$-A$	2	Carbon starvation marker (ASN1/DIN6 induction)
L-Valine	C00183	$-A$	2	Branched-chain amino acid mobilized during starvation
L-Leucine	C00123	$-A$	2	Branched-chain amino acid catabolized for energy
L-Isoleucine	C00407	$-A$	2	Branched-chain amino acid catabolized for energy
L-Glutamine	C00064	$-A$	2	Nitrogen remobilization hub during carbon deficit
L-Glutamate	C00025	$-A$	2	Amino donor for transamination and GS/GOGAT hub

Supplementary Figure S1 — transcript, protein and pathway together



Ribbon width is a locus count. Columns 1 and 2 conserve loci exactly; column 3 counts memberships. Three whole-ontology MapMan maps are excluded (each covers more than 25% of measured loci). Blue square = KEGG, green = MapMan, as in Fig. 6a.

Fig. S1 Where the loci go, and how far the MapMan half of the diagram can be trusted. **a**, Each locus in at least one of the 29 non-whole-ontology significant pathways is placed in one transcript bin and one protein bin, and then into the pathways it belongs to. Ribbon width is a locus count. The transcript and protein columns conserve loci exactly; the pathway column counts *memberships*, so a locus in more than one pathway is counted more than once. The pathway column draws the ten most significant; a square gives each pathway's database in the colours of Fig. 6a. Three whole-ontology MapMan diagrams are excluded — *Overview* lists all 36 top-level bins and so covers 99.8 % of measured loci, and *Regulation overview* and *Biotic Stress* cover 45 % and 44 %; the next largest covers 17 %. A map holding nearly every measured locus cannot say where anything goes, because its ribbons reproduce the marginal distribution. **b**, PaintOmics reports enrichment but not the gene lists behind it, and MapMan has no equivalent of the KEGG REST API, so MapMan membership is reconstructed from GoMapMan's Araport11 gene-to-bin mapping and the MapMan diagram layouts, by PaintOmics' own inclusion rules. Reconstructed membership is compared against the feature count PaintOmics itself reported for each diagram (log scale). Nine of the thirteen agree to within three features. Significance throughout is PaintOmics' uncorrected combined $p < 0.05$. The four that do not are the largest maps, are marked in vermilion with the ratio between the two counts, and the reason for the difference is not recoverable from PaintOmics' published sources; both counts are reported rather than one.